

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

Intensity Frequency

2	4
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

[10 10 10, 10 20 10, 10 10 10]

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$\begin{bmatrix} -1 & -2 & -1, & 0 & 0 & 0, & 1 & 2 & 1 \end{bmatrix}$

to:

[10 10 10, 20 20 20, 30 30 30]

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0	4
1	6
2	10
3	6

- (a) Interpret the need for equalization.
- (b) Justify use of CDF.
- (c) Compute cumulative distribution.
- (d) Compute new intensity levels.
- (e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

- (a) Interpret noise characteristics.
- (b) Justify median filter usage.
- (c) Apply filter to:

10 200 10,200 50 200,10 200 10

- (d) Compute center value.
- (e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

- (a) Interpret smoothing objective.
- (b) Justify Gaussian filter over averaging.
- (c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10,20 40 20,10 20 10]

- (d) Compute center pixel.
- (e) Interpret smoothing effect.

16. High-Pass Filtering

An image requires sharpening.

- (a) Interpret need for high-pass filtering.

- (b) Justify filter selection.
- (c) Apply mask:

$$\begin{bmatrix} -1 & -1 & -1, -1 & 8 & -1, -1 & -1 & -1 \end{bmatrix}$$
- (d) Compute center value.
- (e) Interpret edge enhancement.

17. Frequency Domain High-Pass

Given frequency components:

$$F = [10, 20, 80, 100]$$

- (a) Interpret high-frequency importance.
- (b) Justify high-pass filtering.
- (c) Apply filtering (remove low frequencies).
- (d) Compute output.
- (e) Interpret result.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

- (a) Interpret segmentation challenge.
- (b) Justify adaptive thresholding.
- (c) Apply local threshold (mean = 100).
- (d) Compute binary output.
- (e) Interpret result.

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$\begin{bmatrix} -1 & 0 & 1, -2 & 0 & 2, -1 & 0 & 1 \end{bmatrix}$$

Matrix:

$$\begin{bmatrix} 10 & 20 & 30, 10 & 20 & 30, 10 & 20 & 30 \end{bmatrix}$$

- (a) Interpret edge detection objective.
- (b) Justify operator choice.
- (c) Apply step-by-step.
- (d) Compute center value.
- (e) Interpret edge direction.

20. Prewitt Edge Detection

- (a) Interpret Prewitt operator purpose.
- (b) Justify use over Sobel.
- (c) Apply kernel:

$$\begin{bmatrix} -1 & 0 & 1, -1 & 0 & 1, -1 & 0 & 1 \end{bmatrix}$$
- (d) Compute output.
- (e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

- (a) Interpret segmentation problem.
- (b) Justify splitting method.
- (c) Apply threshold 80.
- (d) Compute regions.
- (e) Interpret separation.

22. Region Merging

Regions: [45, 50], [52, 55]

- (a) Interpret merging criteria.
- (b) Justify merging approach.
- (c) Apply threshold = 5.
- (d) Compute merged region.
- (e) Interpret result.

23. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

- (a) Interpret problem of discontinuity.
- (b) Justify linking method.
- (c) Apply linking step.
- (d) Compute result.
- (e) Interpret continuity.

24. Low-Pass Filtering (Frequency)

Given $F = [200, 150, 50, 20]$

- (a) Interpret smoothing requirement.

- (b) Justify LPF selection.
- (c) Apply filter.
- (d) Compute output.
- (e) Interpret effect.

25. Laplacian Sharpening (Variant)

Mask:

$$\begin{array}{ccc} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{array}$$

- (a) Interpret sharpening need.
- (b) Justify mask.
- (c) Apply to matrix.
- (d) Compute center.
- (e) Interpret result.

26. Histogram Matching

- (a) Interpret need for matching.
- (b) Justify method.
- (c) Map values [10, 50, 100].
- (d) Compute new values.
- (e) Interpret.

27. Noise Reduction Comparison

- (a) Interpret noise type.
- (b) Compare averaging vs median.
- (c) Apply both.
- (d) Compute outputs.
- (e) Interpret differences.

28. Combined Filtering

- (a) Interpret need for smoothing + sharpening.
- (b) Justify sequence.
- (c) Apply filters stepwise.

- (d) Compute output.
- (e) Interpret improvement.

29. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

- (a) Interpret multi-level segmentation.
- (b) Justify thresholds (50,100).
- (c) Apply segmentation.
- (d) Compute classes.
- (e) Interpret.

30. Boundary Detection

- (a) Interpret boundary extraction.
- (b) Justify method.
- (c) Apply edge + linking.
- (d) Compute result.
- (e) Interpret boundaries.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1) Grey Level Transformation

a) Apply the purpose of grey level transformation in image enhancement

b) Apply a negative transformation $s = 255 - r$ to pixel values: $r = [50, 100, 150, 200]$
Compute the transformed values.Ans) a) * Improves image brightness and contrast
* Enhances image quality.b) Formula: $s = 255 - r$

r	s
50	205
100	155
150	105
200	55

2) Log Transformation:

a) Explain how log transformation enhances low intensity pixels

b) Given $s = c \log(1+r)$, where $c = 10$, compute s for $r = [0, 10, 50, 100]$.

Ans: a) * Enhances dark pixels and compresses bright pixels

b) * Formula: $s = 10 \log(1+r)$

r	s
0	0
10	10.41
50	17.08
100	20.04

- 3) a) * Improves image contrast
* Distributes pixel intensities uniformly

b) CDF:

Intensity	CDF	New intensity
0	2	1
1	4	2
2	8	4
3	16	7

4) Spatial Filtering (Smoothing)

a) Explain the purpose of spatial filtering for smoothing.

b) Apply a 3×3 averaging filter to the following region.

$[10, 20, 30, 20, 30, 40, 30, 40, 50]$

Ans) a) • Removes image noise
• Smooths the image

b) $\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$ Calculation = $\frac{10+20+30+40+50+60+70+80+90}{9} = \frac{450}{9} = 50$
New Center Pixel = 50.

5) Spatial Filtering (Sharpening)

a) Explain how sharpening enhances image details

b) Apply the Laplacian mask to the matrix. Compute the output at the center.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Ans) a) • Enhances edges
• Highlights fine details

b) $\begin{bmatrix} 20 & 20 & 20 \\ 20 & 50 & 20 \\ 20 & 20 & 20 \end{bmatrix} = (4 \times 50) - (20 + 20 + 20 + 20) = 200 - 80 = 120$
 $= 120$

6) Frequency Domain Filtering

- Differentiate low-pass and high-pass filtering in frequency domain.
- Given frequency components, apply a low-pass filter that retains only the first two components. Write the filtered output.

Ans: a) Low-pass Filter

Removes noise

Smooths image

Keeps low frequencies

High-pass filter

Enhances edges

Sharpens image

Keeps high frequencies.

b) Frequency components [10, 20, 30, 40]

Filtered Output [10, 20, 0, 0]

7) Thresholding

a) Explain global thresholding in segmentation

b) ~~Separates~~ Given pixel values, apply threshold $T = 100$. Assign a value (0 or 1).

Ans: a) • Separates object from background
• Uses one threshold value

b) Given pixels [30, 80, 120, 200]

Pixels

Output

30

0

80

0

120

1

200

1

Binary Output = [0 0 1 1]

8) Edge Detection

a) Explain the concept of edge detection

b) Apply the Sobel operator (horizontal). Compute the gradient at the center

Ans

- a) • Detects object boundaries
• Finds sudden intensity changes
• Used in image segmentation.

b) Sobel kernel = $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ Given Matrix = $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Calculation = $(-1 \times 1) + (-2 \times 2) + (-1 \times 1) + (1 \times 1) + 2 \times 2 + (1 \times 1) = 0$

Gradient at Center = 0.

a) Region Based Segmentation.

- a) Explain region growing in segmentation
b) Given pixel intensities, if seed = 50 and threshold difference = 5, identify pixels belonging to the region.

Ans

- a) • starts from a seed pixel
• Groups neighbouring pixels with similar intensity.
• Forms a region.

b) Given pixels = [45, 48, 50, 53, 60]

seed = 50

Threshold Difference = 5

Selected Region = 45, 48, 50, 53

10) Combined Enhancement & Segmentation.

- a) • Removes image noise
• Improves segmentation accuracy
• produces better object detection.

b) Given, [30, 45, 50, 60]

Pixel	Output
30	0
45	0
50	1
60	1

Binary segmentation output = [0, 0, 1, 1]

11) Power Law (Gamma) Transformation.

- a) Gamma correction adjusts image brightness.
- b) used to control brightness

c)

Pixel	$s = \sqrt{r}$
25	5
49	7
64	8
81	9

- c) Brightness increases.

12) Contrast stretching.

- a) improves image contrast
- b) Maps intensity range from 50-150 to 0-255

c)

r	s
50	0
75	64
100	128
150	255

- d) 0, 64, 128, 255

13) Histogram Equalization

- a) Improves image contrast
- b) CDF is used to compute new intensity values

c)

Intensity	0	1	2	3
Frequency	4	6	10	6
CDF	4	10	20	26

- d) • Better contrast

14) Median Filtering.

- a) Removes salt and pepper noise
- b) Median Filter removes noise without blurring
- c) Arrange = 10, 20, 30, 40, 50, 60, 70, 80, 255
- d) Median (center value)
- e) Noise removed.

15) Gaussian smoothing

a) Reduces image noise

b) Gaussian filter gives smoother results

c)
$$= \frac{10+40+30+80+200+120+70+160+90}{16} = \frac{800}{16} = 50$$

d) Noise reduced.

16) High-pass Filtering

a) Enhances Edges

b) It preserves high-frequency components

c)
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

d)
$$(5 \times 50) - (20+20+20+20) = 250 - 80 = 170$$

e) Image becomes sharper.

17) Frequency domain high pass.

a) High frequency components represent edges

b) High-pass filter removes low frequencies

c) $[0, 0, 30, 40]$

d) $[0, 0, 30, 40]$

e) sharp edges appear.

18) Adaptive threshold.

a) Handles uneven lighting

b) uses local threshold

c) $[60, 80, 120, 140]$

d) Pixel 60 80 120 140

output 0 0 1 1

e) Accurate segmentation.

19) Edge Detection (vertical sobel)

a) Detects vertical edges

b) Sobel operator is simple and effective

c)
$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

d)
$$\begin{aligned} &= (-1 \times 1) + (1 \times 3) + (-2 \times 4) + (2 \times 6) + (-1 \times 7) + (1 \times 9) \\ &= -1 + 3 - 8 + 12 - 7 + 9 = 8. \end{aligned}$$

20) Prewitt Edge Detection

- a) Detects image edges
- b) Simpler than sobel
- c)
$$= (-1 \times 1) + (1 \times 3) + (-1 \times 4) + (1 \times 6) + (-1 \times 7) + (1 \times 9)$$
$$= 6$$
- d) Detects vertical edges.

21) Region splitting.

- a) Separates image into different region
- b) Justify splitting method.
- c) 80
- d) Region 1 [40, 50], Region 2 [100, 105]

22) Region Merging.

- a) Interrupt merging criteria
- b) Justify merging approach
- c) Apply threshold $52 - 50 = 2 < 5$
- d) [45, 50, 52, 55]
- e) Similar regions are merged.

23) Edge Linking.

- a) • Edge detection may produce broken edges
- b) Connects nearby edge pixels
- c) [1, 1, 1, 1, 1]
- d) Edge becomes continuous

24) Laplacian Sharpening (variant)

- a) Improves image clarity
- b) Detects rapid intensity changes
- c) $= (8 \times 50) - (10 \times 8) = 460 - 80 = 320$
- d) Image becomes sharper

25) Low-pass Filtering

- a) Removes image noise
- b) Preserves low frequency components
- c) Components [15, 25, 35, 45]
- d) Output [15, 25, 0, 0]
- e) Noise reduced.

- 26) Histogram Matching
- a) Makes one image similar to another.
 - b) Matches histogram with reference image.
 - c) Map Values [10, 50, 100]
 - d) [20, 80, 150]
 - e) Brightness becomes similar to the reference image.

27) Noise Reduction Comparison.

- a) Noise reduces image quality.
- b) Blurs image
- c) [10, 20, 255, 30, 40]
- d) Averaging = 71, Median = 30
- e) Median filter removes noise better.

28) Combined Filtering.

- a) Removes noise
- b) Apply smoothing first
- c) Original $\rightarrow$ smoothing - sharpening.
- d) Original = 100, Sharpening = 110.
- e) Sharper edges

29) Multi-Threshold Segmentation.

- a) Divides image into multiple regions
- b) separates low, medium, and high intensities
- c) Pixels = [20, 60, 120, 180]
- d) Pixel 20 60 120 180
class Low Med High High
- e) Three intensity groups are obtained.

30) Boundary Detection.

- a) Detects object boundaries
- b) uses edge detection and edge linking.
- c) After Linking [1, 1, 1, 1]
- d) Continuous object boundary is obtained.

